Daily

Aujourd'hui, l'équipe Dev a :

- Installer 3 machines debian
- Adresser ces 3 machines
- Installer k3s sur les 3 machines
- Donner les rôles aux différentes machines
- La configuration k3s
- Faire rejoindre les workers au master
- Mettre en place une app-web avec ngninx qui fonctionne

Prochainement nous allons mettre en place l'API ainsi qu'exposer les services

Adressage des machines

Noms des machines	IP	Rôles
KubeCP-GRP-3	172.16.130.60	Master (Control-Plane)
Kubeworker-GRP-3	172.16.130.161	Worker 1
Kubeworker2-GRP-3	172.16.130.162	Worker 2

Etape — Preparer Debian sur les 3 VM

2.1 Mettre a jour le systeme

```
sudo apt update && sudo apt upgrade -y
```

Désactiver le swap (OBLIGATOIRE pour k3s)

k3s refuse de démarrer si le swap est actif.

```
sudo swapoff -a
sudo sed -i '/ swap / s/^/#/' /etc/fstab

# Verifier : Swap doit afficher 0
free -h
```

2.3 Activer le forwarding reseau

```
echo 'net.ipv4.ip_forward = 1' \| sudo tee /etc/sysctl.d/k8s.conf

sudo sysctl --system
```

```
grp3@KubeCP-GRP-3:~$ sudo sysctl --system
* Applying /usr/lib/sysctl.d/10-coredump-debian.conf ...
* Applying /usr/lib/sysctl.d/50-default.conf ...
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
* Applying /etc/sysctl.d/k8s.conf ...
kernel.core_pattern = core
kernel.sysrq = 0x01b6
kernel.core_uses_pid = 1
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.eth0.rp_filter = 2
net.ipv4.conf.lo.rp_filter = 2
net.ipv4.conf.default.accept_source_route = 0
net.ipv4.conf.eth0.accept_source_route = 0
net.ipv4.conf.lo.accept_source_route = 0
net.ipv4.conf.default.promote_secondaries = 1
net.ipv4.conf.eth0.promote_secondaries = 1
net.ipv4.conf.lo.promote_secondaries = 1
net.ipv4.ping_group_range = 0 2147483647
net.core.default_qdisc = fq_codel
fs.protected_hardlinks = 1
fs.protected_symlinks = 1
fs.protected_regular = 2
fs.protected_fifos = 1
vm.max_map_count = 1048576
kernel.pid_max = 4194304
net.ipv4.ip_forward = 1
grp3@KubeCP-GRP-3:~$
```

Etape 3 — Installer k3s

Nous allons installer k3s sur le **master** en premier et ensuite sur les workers

Installer k3s sur le MASTER

```
curl -sfL https://get.k3s.io \| sh -

sudo kubectl get nodes # pour voire les noeuds
```

```
grp3@KubeCP-GRP-3:~$ curl -sfL https://get.k3s.io | sh -
[INFO]  Finding release for channel stable
[INFO]  Using v1.34.5+k3s1 as release
[INFO]  Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.34.5%2Bk3s1/sha256sum-amd64.txt
[INFO]  Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.34.5%2Bk3s1/k3s
[INFO]  Verifying binary download
[INFO]  Installing k3s to /usr/local/bin/k3s
[INFO]  Skipping installation of SELinux RPM
[INFO]  Creating /usr/local/bin/kubectl symlink to k3s
[INFO]  Creating /usr/local/bin/crictl symlink to k3s
[INFO]  Creating /usr/local/bin/ctr symlink to k3s
[INFO]  Creating killall script /usr/local/bin/k3s-killall.sh
[INFO]  Creating uninstall script /usr/local/bin/k3s-uninstall.sh
[INFO]  env: Creating environment file /etc/systemd/system/k3s.service.env
[INFO]  systemd: Creating service file /etc/systemd/system/k3s.service
[INFO]  systemd: Enabling k3s unit
Created symlink '/etc/systemd/system/multi-user.target.wants/k3s.service' → '/etc/systemd/system/k3s.service'.
[INFO]  systemd: Starting k3s
grp3@KubeCP-GRP-3:~$ sudo kubectl get nodes
NAME           STATUS   ROLES           AGE   VERSION
kubecp-grp-3   Ready    control-plane   15s   v1.34.5+k3s1
grp3@KubeCP-GRP-3:~$
```

Installer k3s sur les WORKERS

Récupérer le token du master

```
sudo cat /var/lib/rancher/k3s/server/node-token
## Le résultat obtenu sera copié
```

Sur k3s-worker1

```
curl -sfL https://get.k3s.io | K3S_URL=https://172.16.130.60:6443 \
K3S_TOKEN=K10f69f79b990822df021ce2391bc6812b4ab227a2b9f14c909a7fecd9549a0b0d0:
:server:30aa662e2bc9fdc846c3f1ce16c6e322 sh -
```

```
grp3@Kubeworker-GRP-3:~$ curl -sfL https://get.k3s.io | K3S_URL=https://172.16.130.60:6443   K3S_TOKEN=K10f69f79b990822df021ce2391bc6812b4ab227a
2b9f14c909a7fecd9549a0b0d0::server:30aa662e2bc9fdc846c3f1ce16c6e322 sh -
[INFO]  Finding release for channel stable
[INFO]  Using v1.34.5+k3s1 as release
[INFO]  Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.34.5%2Bk3s1/sha256sum-amd64.txt
[INFO]  Skipping binary downloaded, installed k3s matches hash
[INFO]  Skipping installation of SELinux RPM
[INFO]  Skipping /usr/local/bin/kubectl symlink to k3s, already exists
[INFO]  Skipping /usr/local/bin/crictl symlink to k3s, already exists
[INFO]  Skipping /usr/local/bin/ctr symlink to k3s, already exists
[INFO]  Creating killall script /usr/local/bin/k3s-killall.sh
[INFO]  Creating uninstall script /usr/local/bin/k3s-agent-uninstall.sh
[INFO]  env: Creating environment file /etc/systemd/system/k3s-agent.service.env
[INFO]  systemd: Creating service file /etc/systemd/system/k3s-agent.service
[INFO]  systemd: Enabling k3s-agent unit
Created symlink '/etc/systemd/system/multi-user.target.wants/k3s-agent.service' → '/etc/systemd/system/k3s-agent.service'.
[INFO]  No change detected so skipping service start
grp3@Kubeworker-GRP-3:~$
```

Sur k3s-worker2 :

```
curl -sfL https://get.k3s.io | K3S_URL=https://172.16.130.60:6443 \
K3S_TOKEN=K10f69f79b990822df021ce2391bc6812b4ab227a2b9f14c909a7fecd9549a0b0d0:
:server:30aa662e2bc9fdc846c3f1ce16c6e322 sh -
```

```
grp3@Kubeworker2-GRP-3:~$  curl -sfL https://get.k3s.io | K3S_URL=https://172.16.130.60:6443   K3S_TOKEN=K10f69f79b990822df021ce2391bc6812b4ab227a2b9f14c909
a7fecd9549a0b0d0::server:30aa662e2bc9fdc846c3f1ce16c6e322 sh -
[INFO]  Finding release for channel stable
[INFO]  Using v1.34.5+k3s1 as release
[INFO]  Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.34.5%2Bk3s1/sha256sum-amd64.txt
[INFO]  Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.34.5%2Bk3s1/k3s
[INFO]  Verifying binary download
[INFO]  Installing k3s to /usr/local/bin/k3s
[INFO]  Skipping installation of SELinux RPM
[INFO]  Creating /usr/local/bin/kubectl symlink to k3s
[INFO]  Creating /usr/local/bin/crictl symlink to k3s
[INFO]  Creating /usr/local/bin/ctr symlink to k3s
[INFO]  Creating killall script /usr/local/bin/k3s-killall.sh
[INFO]  Creating uninstall script /usr/local/bin/k3s-agent-uninstall.sh
[INFO]  env: Creating environment file /etc/systemd/system/k3s-agent.service.env
[INFO]  systemd: Creating service file /etc/systemd/system/k3s-agent.service
[INFO]  systemd: Enabling k3s-agent unit
Created symlink '/etc/systemd/system/multi-user.target.wants/k3s-agent.service' → '/etc/systemd/system/k3s-agent.service'.
[INFO]  systemd: Starting k3s-agent
grp3@Kubeworker2-GRP-3:~$
```

Verifier que les 3 noeuds sont Ready

Sur le **Master**

```
kubectl get nodes
```

```
grp3@KubeCP-GRP-3:~$ sudo kubectl get nodes
NAME                STATUS   ROLES           AGE    VERSION
kubecp-grp-3        Ready    control-plane   35m    v1.34.5+k3s1
kubeworker-grp-3    Ready    <none>          117s   v1.34.5+k3s1
kubeworker2-grp-3   Ready    <none>          26s    v1.34.5+k3s1
grp3@KubeCP-GRP-3:~$
```

Etape — Deployer les 3 applications

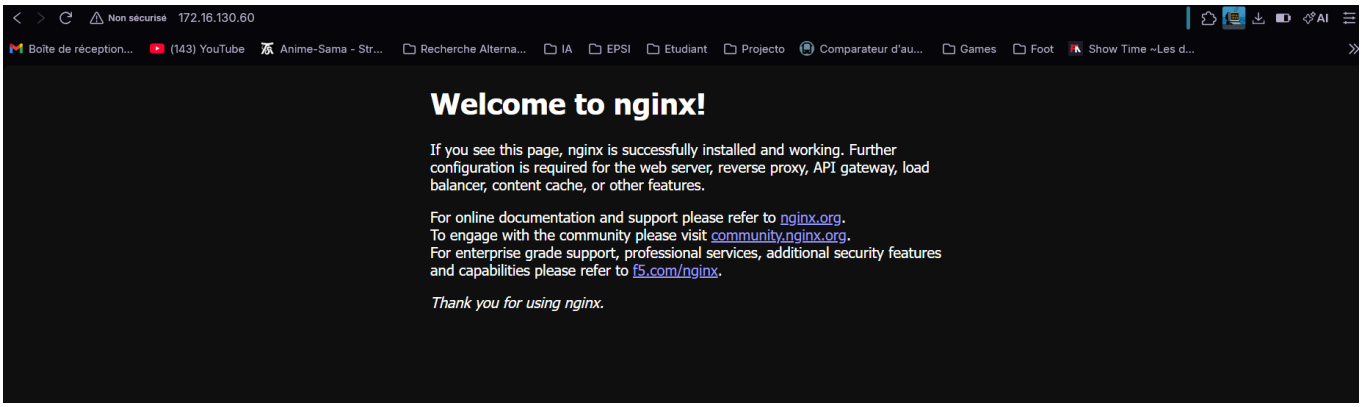
Sur le **master**

```
sudo mkdir appli
cd appli

sudo nano app-web.yml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: app-web
spec:
  replicas: 2
  selector:
    matchLabels:
      app: app-web
  template:
```

```
    metadata:
      labels:
        app: app-web
    spec:
      containers:
      - name: app-web
        image: nginx:alpine
        ports:
        - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: app-web-svc
spec:
  selector:
    app: app-web
  ports:
  - port: 80
    targetPort: 80
```

Pour appliquer le fichier il faut :

```
kubectl apply -f app-web.yaml
```